MUSE: trustworthy AI when medical scans are missing

Modality-Universal Selective Evidence Fusion — the full story: problem, ideas, code, results

The story in one slide

1 · Problem

Brain-tumour AI needs 4 MRI scans, but patients often miss some. Current AI fakes them or breaks.

2 · Idea

Treat each scan as an expert giving evidence. A missing scan simply says nothing — no faking.

3 · Build

One 3D network per scan, evidence fusion, uncertainty and explanations, tested on 98 real patients.

4 · Result

Matches the best published-style methods on accuracy, and adds trustworthy uncertainty and explanations.

Chapter 1 · The problem

Why missing scans break today's medical AI

A brain-tumour patient needs four kinds of MRI scan

FLAIR

Highlights swelling (edema) around the tumour.

T1

Shows normal anatomy clearly.

T1gd

T1 after contrast dye: lights up the active, growing tumour.

T2

Fluid-sensitive; shows tumour and swelling together.

The AI's job: outline the tumour and its parts on every 3D pixel (voxel) of the brain.

In real hospitals, scans are often missing

15

possible combinations of available scans

How today's AI copes — and why it fails

- Fake the missing scan with a generative AI — the model then trusts invented data
- Fill it with zeros — the network never saw that in training and fails silently
- Train 15 separate models — impractical to build, validate and deploy
- Drop the patient — the patients who most need help are excluded

Chapter 2 · The vocabulary

Every technical word, in plain language

Words about images and results

Segmentation
Colouring every voxel by what it is (healthy, swelling, tumour core, active tumour).

Voxel
A 3D pixel — here 1 mm × 1 mm × 1 mm of brain.

WT / TC / ET
Whole tumour, tumour core, enhancing (active) tumour — the three regions doctors care about.

Dice score
Overlap between AI outline and expert outline: 0 = none, 1 = perfect.

HD95
How far (mm) the AI's border strays from the true border; lower is better.

Missing modality
A scan type that was not acquired for this patient.

Words about neural networks

Neural network
A program that learns patterns from examples instead of hand-written rules.

3D U-Net
The standard medical-image network: shrinks the scan to see context, then expands back to full detail.

Encoder / decoder
The shrinking half (understands) and the expanding half (draws the outline).

Training / validation / test
Learn on one set, choose settings on a second, report on a third never seen before.

Seed
The random starting point of training; repeating with several seeds shows results are not luck.

Graph neural network
A network for data connected like a web (papers citing papers); used for MUSE's second test bed.

Words about evidence and uncertainty

Evidence (α)
How strongly a scan supports each class. α = 1 means 'no evidence at all'.

Dirichlet distribution
The maths that turns evidence counts into probabilities plus uncertainty.

Uncertainty u = C / S
Number of classes ÷ total evidence: little evidence → high uncertainty.

Dempster–Shafer theory
A theory of evidence that can say 'I don't know' instead of forcing a guess.

Vacuous mass
The formal 'I know nothing' opinion — exactly what a missing scan contributes.

Cumulative fusion
Combine opinions by adding their evidence (from subjective logic).

Words about judging results fairly

Baseline
An existing method we compare against (HeMIS, RFNet, mmFormer, …).

Ablation
Remove one part of MUSE and re-test, to prove that part actually helps.

Calibration (ECE)
Does 90% confidence mean right 90% of the time? Lower error = more honest.

AUROC (error detection)
How well the uncertainty points at the AI's own mistakes; 1 = perfectly.

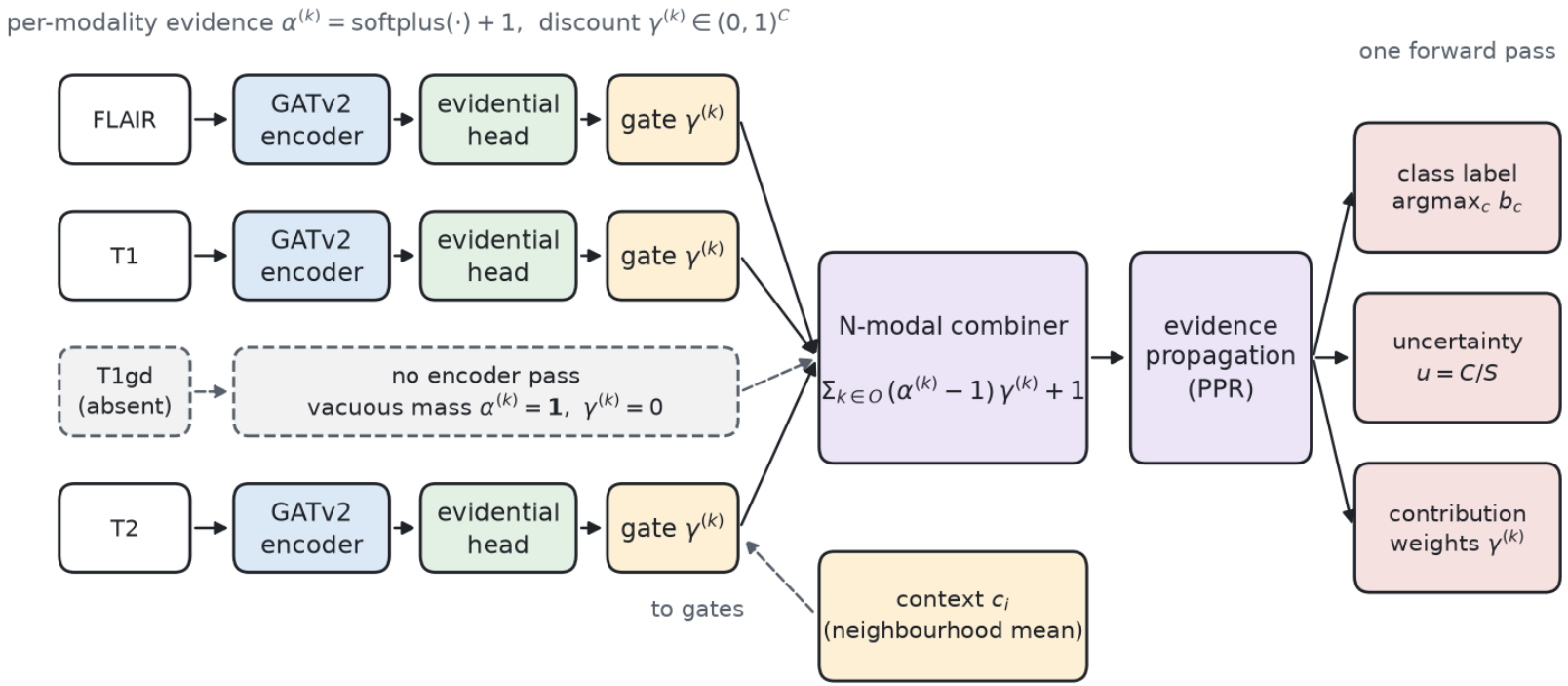
p-value (Wilcoxon test)
Chance the difference is luck; below 0.05 counts as significant.

Pull request (PR)
A reviewed, recorded change to the code on GitHub — our audit trail.

Chapter 3 · How MUSE works

A panel of experts that never fakes evidence

MUSE architecture: one forward pass, three outputs



- Own network per scan
- Missing scan = vacuous mass
- Gate = trust per class & voxel
- Fusion = add the evidence
- Outputs: outline, uncertainty, who-said-what

The same idea as a panel of four experts

Each expert reads one scan

FLAIR expert, T1 expert, … each marks where it thinks the tumour is.

Missing scan → expert stays silent

No guessing, no invented data. Silence adds exactly nothing.

Votes are weighted by trust

The contrast-scan expert counts more for active tumour — MUSE learns this.

Weighted votes are added up

Agreement → confident. Little evidence or conflict → uncertain, and MUSE says so.

Three guarantees MUSE can prove

Order never matters

Fusing FLAIR then T2, or T2 then FLAIR, gives the identical answer (checked: difference 0).

Missing scans are harmless

A missing scan adds exactly zero — one model serves all 15 combinations.

More scans never raise doubt

Adding a scan can only lower uncertainty (checked on real patients: 0 violations).

Honest note: these follow from known 'cumulative fusion' maths; MUSE's novelty is the complete trustworthy system built on them.

Chapter 4 · The reasoning journey

What we changed, and why — decided on validation data only

First, we corrected the original plan

False claim removed

'Dempster's rule is order-dependent' — it is not. We proved and measured it: difference 3×10^{-16}.

Wrong sign fixed

The 'balance' term pushed scans apart instead of balancing them. Fixed; the original is kept as an ablation.

Honest novelty

Evidence-sum fusion already exists; MUSE's contribution is the system: gate, vacuous mass, uncertainty, explanations.

Each design step was earned on validation data

- Bug found: MUSE could not learn on Amazon-Photo → added normalisation
- MRI as coarse graphs capped accuracy → rebuilt as full 3D U-Nets
- Same fixes given to baselines that broke (ECD, EGNN) — fairness

Chapter 5 · How the code is built

From raw scans to tables and figures, reproducibly

The pipeline, end to end

1
Download
Public MSD brain-tumour MRI, 484 patients

2
Preprocess
Crop to brain, normalise each scan

3
Train
3D crops, random scans hidden, 12,000 steps

4
Validate
Pick best checkpoint on unseen validation patients

5
Test
98 patients × 15 scan combinations

6
Report
Tables, significance tests, real-scan figures

Every number in the papers is produced by this code from saved result files — never typed by hand.

How the code is organised

`muse3d/models.py`	MUSE and every MRI baseline on one shared 3D U-Net backbone
`muse3d/losses.py`	Evidence loss + Dice loss; each model's own training objective
`run_mri3d.py`	Preprocess → train (resumable) → test all 15 combinations
`muse/ & run_graph.py`	Graph version of MUSE and the citation / product benchmarks
`aggregate*.py, figures/`	Tables with significance tests, and every figure
`tests/`	14 automatic checks, e.g. a missing scan can never change the output

How we kept it rigorous

Tested code

14 automatic tests: properties hold, missing scans have zero effect, fast paths equal slow paths.

No peeking at test data

All choices on validation data; test results computed once at the end.

Every change reviewed

Feature branch → pull request → review → merge: a complete audit trail on GitHub.

Compute where it fit

Mac for testing, free Colab and Kaggle GPUs for training, resumable from checkpoints.

Chapter 6 · What we found

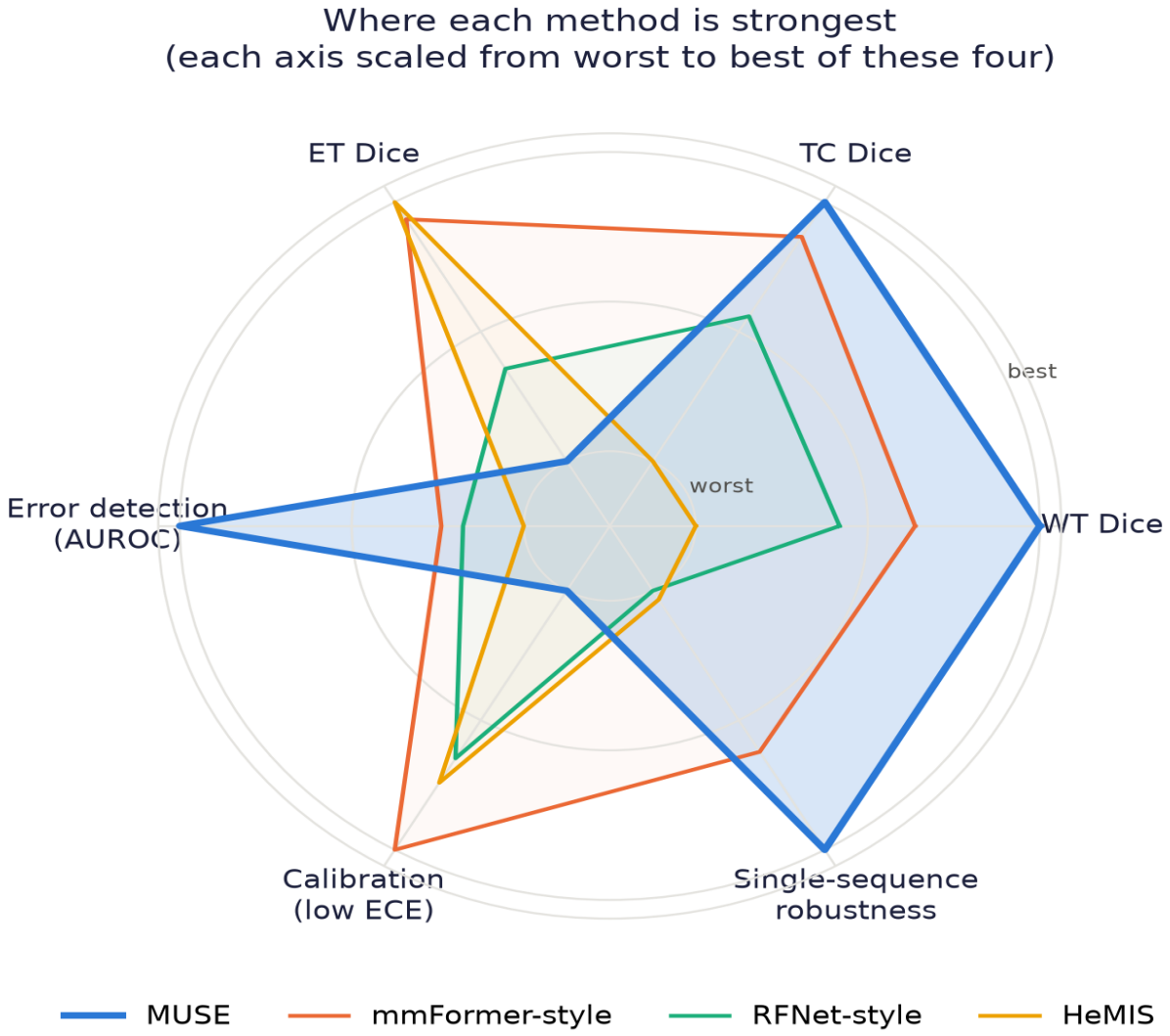
98 unseen patients, 15 scan combinations, 7 methods

Brain-tumour MRI: overall accuracy

Tie at the top

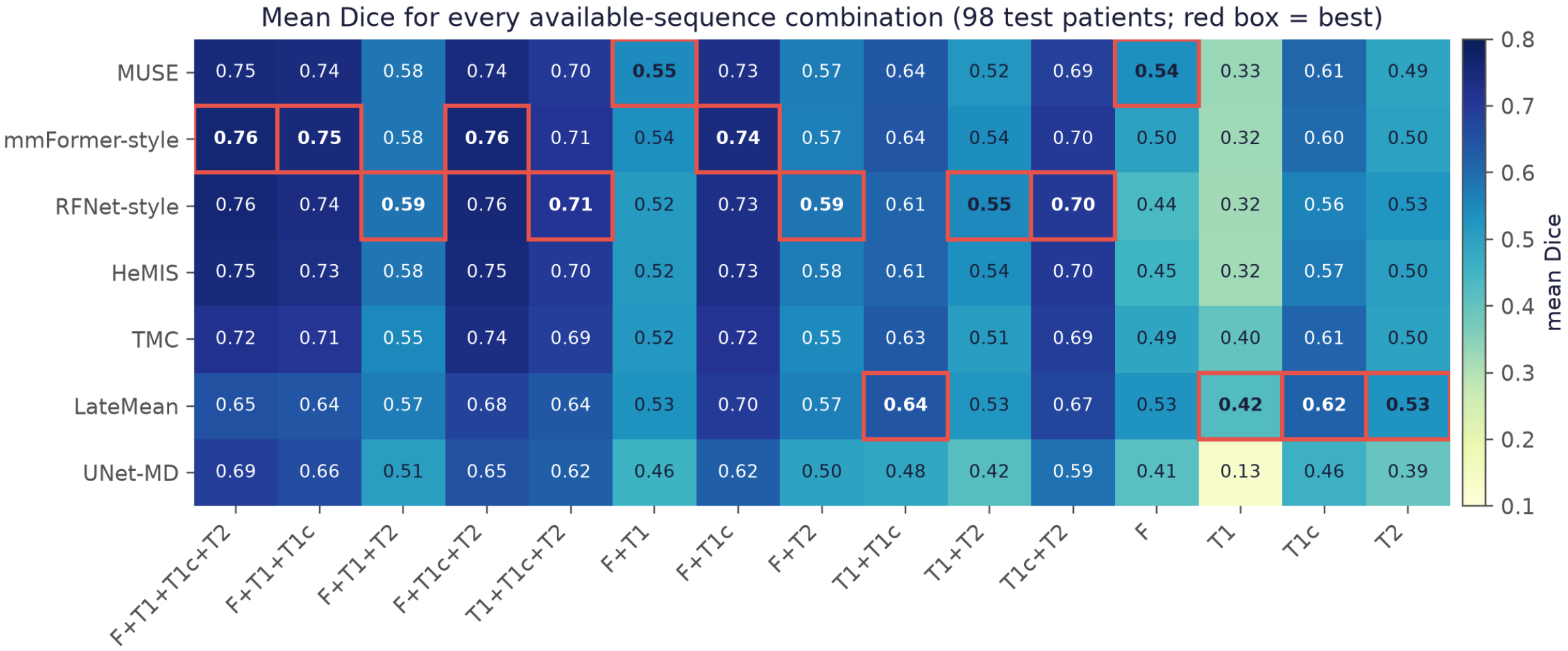
MUSE matches the strongest specialised methods and significantly beats HeMIS, LateMean and the standard U-Net.

Each method's strengths at a glance

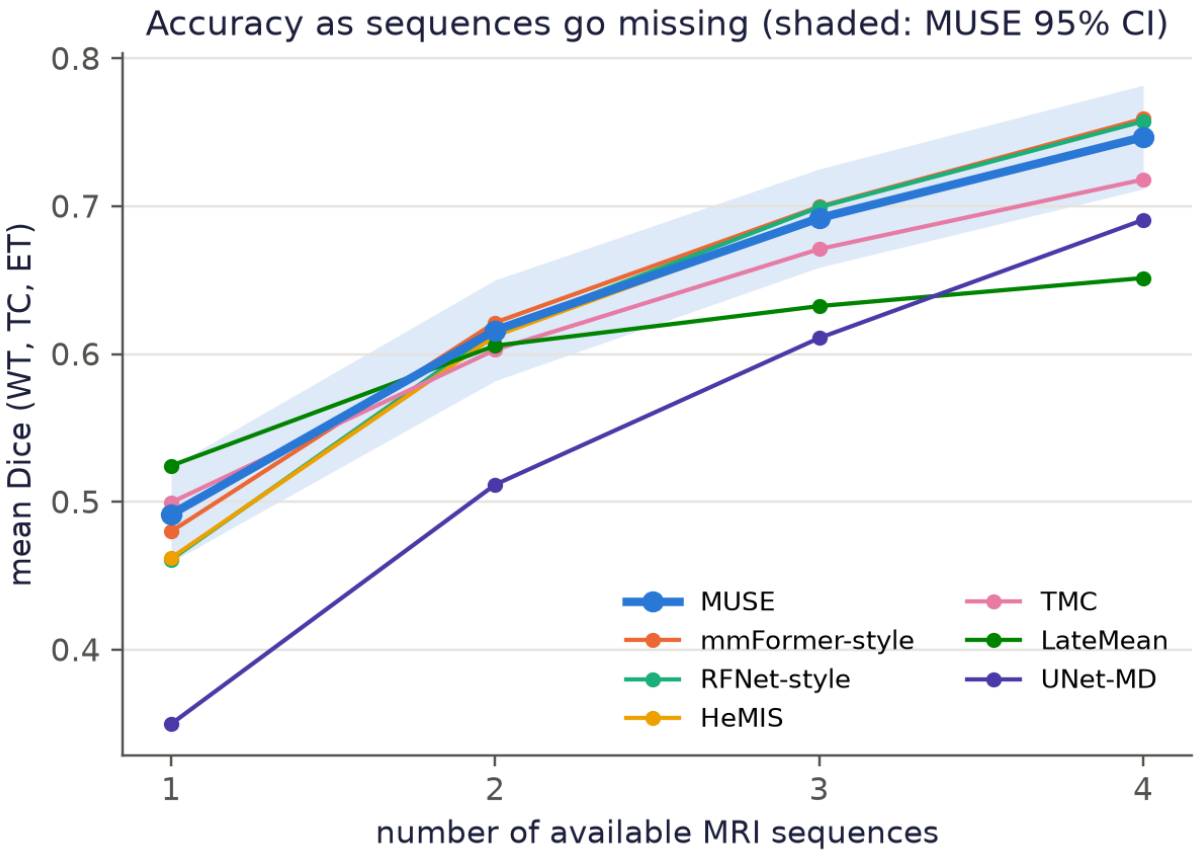


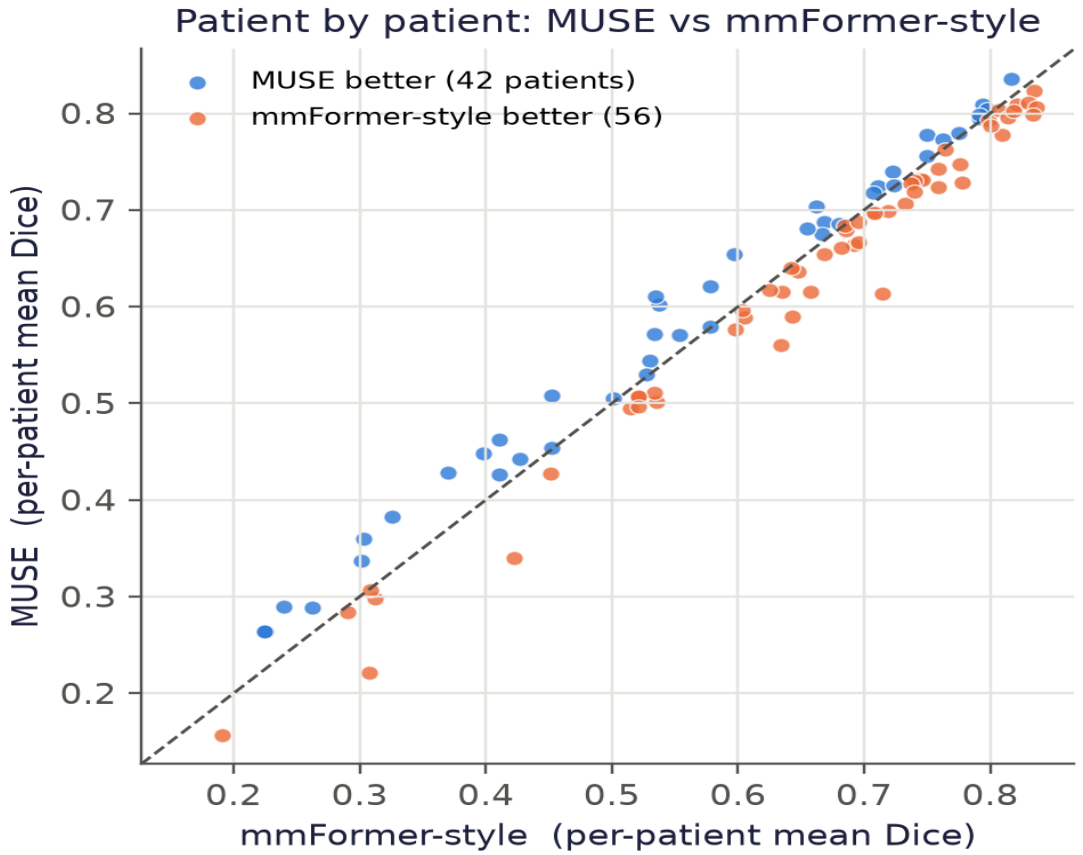
- MUSE leads: whole tumour, tumour core, error detection, single-scan robustness
- mmFormer-style leads: enhancing tumour, calibration
- Axes scaled worst → best of these four, so gaps look larger than they are

Every scan combination, every method



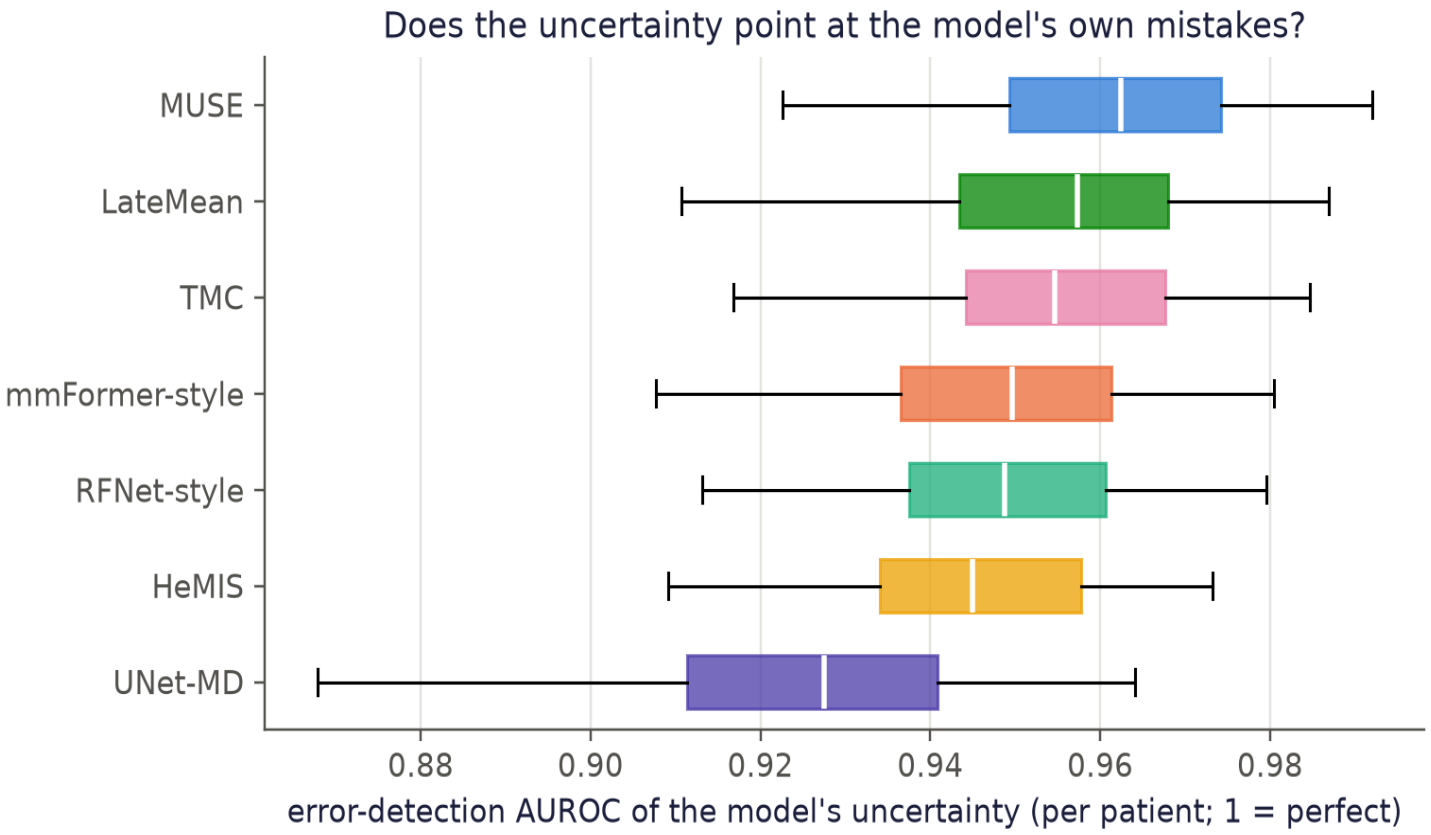
MUSE helps most where segmentation is hardest

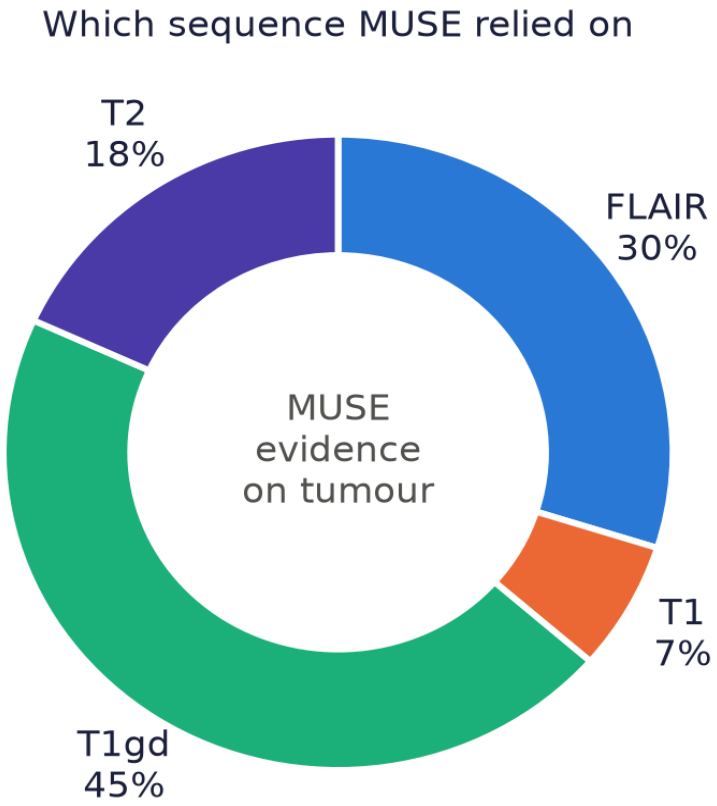




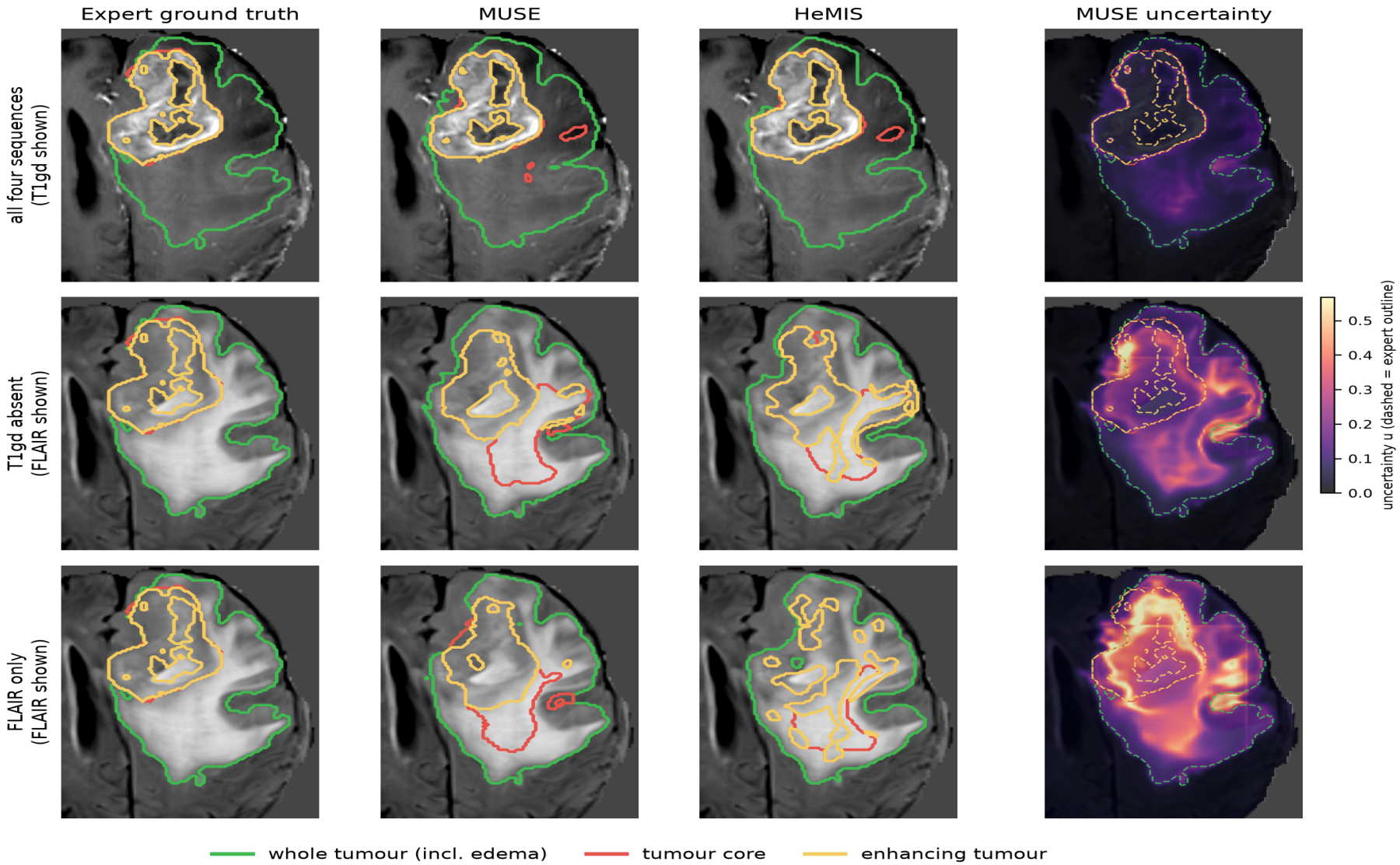
On hard patients (lower-left), MUSE is mostly above the line.

MUSE knows when it is wrong — and shows why





A real unseen patient, scans removed one by one



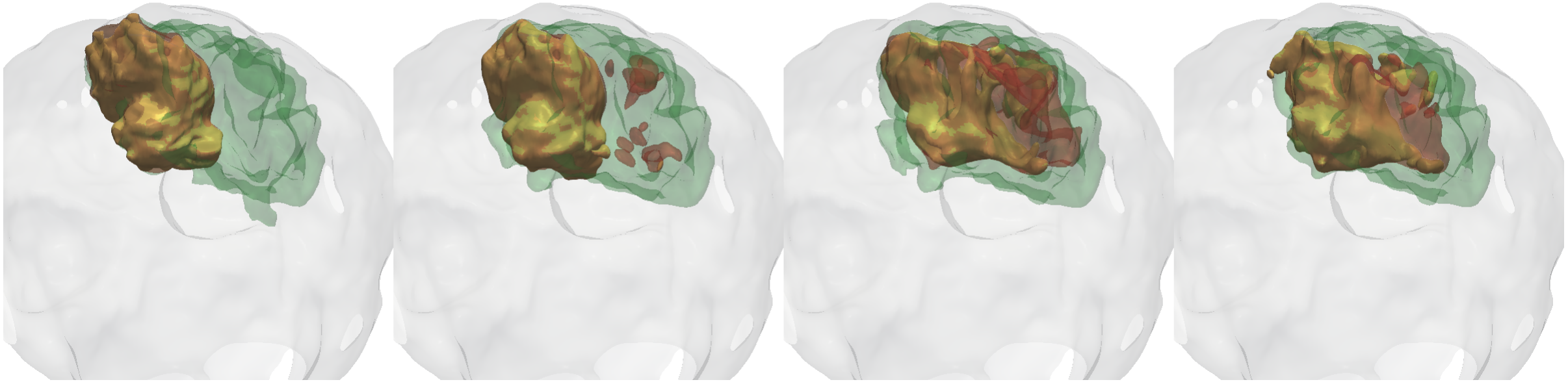
The tumour in 3D: expert vs MUSE

Expert ground truth

MUSE · all four sequences

MUSE · T1gd absent

MUSE · FLAIR only



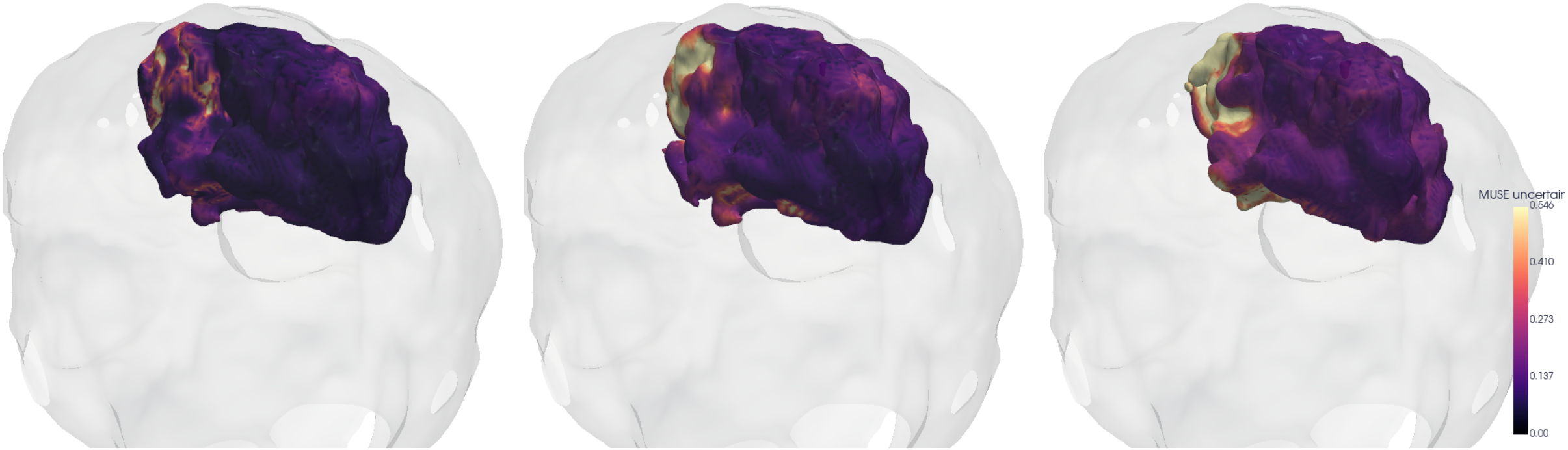
With all four scans MUSE reproduces the expert's tumour shape; with fewer scans it degrades gracefully instead of breaking.

MUSE's doubt, painted on the tumour

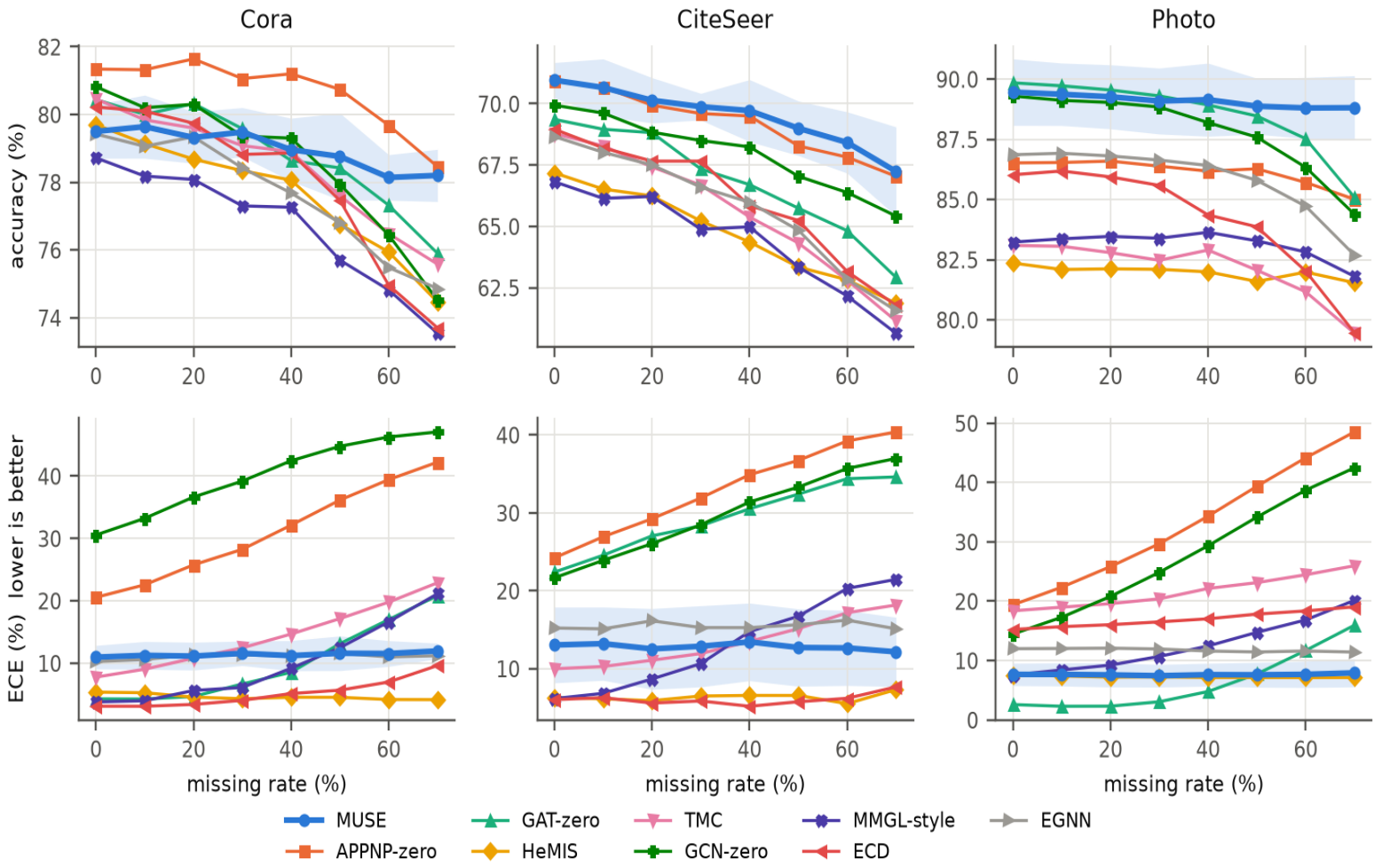
MUSE uncertainty · all four sequences

MUSE uncertainty · T1gd absent

MUSE uncertainty · FLAIR only



The same idea works beyond MRI: graph data



CiteSeer **69.5%**

best of 9

Amazon-Photo **89.1%**

best of 9

Cora **79.0%**

2nd (APPNP 80.7%)

Ablations prove each part earns its place

- Removing the gate costs 2.6 points ($p = 6\times10^{-11}$)
- Dempster's rule instead of sum costs 0.8 points ($p = 0.009$)
- Fusion order: MUSE and Dempster give identical answers in any order; Yager, PCR5 and averaging do not

Chapter 7 · Where we are

Status, the open question, and the papers

Status and next steps

Done

All experiments: 7 MRI methods × 98 patients, 9 graph methods × 3 datasets × 5 seeds, ablations, real-scan figures.

Running now

MUSE2: lets scans share context inside the network, aiming to pull clearly ahead of mmFormer. Kept only if it wins on validation.

Papers

IEEE Transactions journal (method + theory + MRI + graphs) and CARES 2027 (MRI + explainability).

Reviewers may ask

More training seeds for MRI, and published RFNet/mmFormer numbers on BraTS — we will address both.

MUSE never fakes a missing scan

It says what it knows, how sure it is, and which scan told it — at the accuracy of the best specialised methods.